RAJVEER SINGH PALL

Curriculum Vitae · October 2026

Jabalpur, Madhya Pradesh, India · rajveerpall04@gmail.com
rajveer-research.vercel.app · github.com/Rajveer-code · Google Scholar · ORCID 0009-0001-6762-6134

RESEARCH INTERESTS

Trustworthy ML · distribution shift · fairness and calibration · ML and LLM evaluation · causal inference

Common thread: whether a reported performance number still holds once the population, the subgroup or the scoring rule changes.

EDUCATION

B.Tech, Computer Science and Business Systems 2023 – 2027 (expected)
Gyan Ganga Institute of Technology and Sciences, Jabalpur, India
CGPA 7.72 / 10 after six semesters (RGPV); class rank 2nd in Semester VI and 5th in Semester V

SELECTED RESEARCH

Published

Comprehensive Evaluation of Machine Learning for Type 2 Diabetes Risk Prediction: Large-Scale External Validation and Fairness Analysis
R. S. Pall, S. Yadav, S. Bhalerao, S. Sahu, R. Ahluwalia, B. Awadhiya. *IEEE CIPHER-2026* (IEEE Xplore).
DOI 10.1109/CIPHER70417.2026.11523789 · arXiv 2607.16253
Trained an XGBoost risk model on NHANES and validated it externally on BRFSS (1.28 million respondents). AUC fell from 0.794 to 0.717 and was lowest for adults aged 60 and over (0.607). Reported subgroup fairness and SHAP interpretability following TRIPOD-AI.

Submitted and under review

Higher AUC, Fewer Cases Flagged: Subgroup Fairness Metrics Can Reverse at the Decision Threshold in Federated Diabetes Screening
R. S. Pall. Submitted, *ML4H 2026* Findings track (non-archival); decision pending. Code
Federated training raised per-group AUC and closed the White–Black AUC gap, yet at a fixed screening capacity the same models flagged fewer Black respondents. Differences between sites in BMI and age reproduced the reversal (10 seeds, composition-matched centralised control).

Which Leaderboard Orderings Survive a Human Reader?
R. S. Pall. Submitted, *Journal of Data-centric Machine Learning Research (DMLR)*. Code · Leaderboard
Which pairwise model orderings on a public LLM leaderboard (MATH-Hard, 27 models) stay identified when the scorer, not the model, is the uncertain part? A 400-item human audit plus an explicit assumption about how models differ gives a curve, not a single number.

Persistent Racial Disparities in U.S. Mortgage Approval: Evidence from 42 Million Applications, 2020–2024
R. S. Pall. Under review, *Journal of Housing Economics*. SSRN 6334459
Reweighting, within-lender fixed effects, a regression-discontinuity design at 80% LTV and partial-identification bounds on HMDA data. After reweighting on income, loan amount, property value, LTV and debt-to-income ratio, 68% of the 14.95-point raw Black–White approval gap is unexplained.

Preprints and manuscripts

Confidently Wrong: Ranking Inversion in Cross-Network Denial-of-Service Detection
R. S. Pall. Manuscript.
Four model families trained on each of four DDoS testbeds and tested on the others (240 evaluations): 108 ranked attack traffic below benign. Divergence monitors would not have flagged the failures; the models' own confidence did, imperfectly.

IndiaFinBench: An Evaluation Benchmark for Large Language Model Performance on Indian Financial Regulatory Text
R. S. Pall. arXiv preprint. arXiv 2604.19298 · Dataset · Demo
406 expert-annotated question–answer pairs over 192 SEBI and RBI documents, used to evaluate 12 LLMs.

RESEARCH SOFTWARE AND DATA

fairscope Subgroup-stratified, calibration-aware fairness auditing for ML models (v0.3.0). PyPI · GitHub

lm-evaluation-harness issue #4007 Census showing 2,936 of 4,524 generative tasks score an unparseable response as a wrong answer; open upstream discussion. GitHub

TrustShift benchmark Prediction files and result tables for a four-domain deployment-shift audit, with a script that regenerates every table; MIT licence. GitHub · Hugging Face

leaderboard-orderings Verifier audit, human labels, results dataset and public leaderboard for the scorer-audit work. GitHub · Results

IndiaFinBench Dataset and live demo for LLM evaluation on Indian regulatory text. Hugging Face · GitHub

EXPERIENCE

Data Science Intern, Unified Mentor Pvt. Ltd. Apr – Jul 2025
Built ML models for client deliverables: data analysis, feature engineering and SHAP-based interpretability.

Python and Web Development Intern, InternPe Sep – Oct 2024
Python programming and web-development tasks, including backend Flask APIs.

ACADEMIC SERVICE

Program Committee reviewer, W-NUT 2026 (EMNLP 2026 workshop on noisy user-generated text) 2026
Program Committee reviewer, FinNLP 2026 (EMNLP 2026 workshop on financial NLP) 2026

TECHNICAL SKILLS

Languages: Python, SQL, LaTeX
Machine learning: PyTorch, scikit-learn, XGBoost, SHAP, Hugging Face Transformers, lm-evaluation-harness
Statistics and causal inference: calibration (ECE), DeLong and bootstrap confidence intervals, partial identification, EconML (causal forests, double machine learning)
Federated and privacy: Flower, Opacus
Tools: Git, Docker